Computer Methods and Programs in Biomedicine

A Trimodal Feature Fusion Framework for Explainable Brain Tumour Classification Using MRI Images

--Manuscript Draft--

Manuscript Number:	CMPB-D-26-00877
Article Type:	Full Length Article
Section/Category:	Classification and Interpretation
Keywords:	Brain tumour classification; Magnetic resonance imaging (MRI); Trimodal feature fusion; deep learning; Explainable artificial intelligence; Grad-CAM
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Manuscript Region of Origin:	Asia Pacific
Abstract:	Accurate classification of brain tumours from magnetic resonance images is critical for computer-aided diagnosis, enabling precise tumour typing that supports timely clinical assessment and treatment decisions. The challenge lies in the substantial heterogeneity of tumours and the scarcity of high quality, well annotated medical imaging datasets. Most state of the art approaches employ deep learning models, particularly convolutional neural networks and their variants, typically organised as single or dual stream systems optimised for classification accuracy. Many existing methods also rely on unconstrained post-hoc visualisations for explainability, which can yield ambiguous or clinically unreliable interpretations. In this work, we introduce a trimodal feature fusion framework for explainable brain tumour classification that combines a ResNet50-based CNN, a hierarchical transformer, and a lightweight vision transformer. A multi-phase training strategy with exponential moving average parameter aggregation is employed to enhance training stability and generalisation. Model interpretability is achieved through a structured Grad-CAM-based explanation protocol that quantifies activation strength while explicitly disclaiming any claim to tumour localisation or segmentation. We evaluate the framework using 70/30 and 80/20 train–test splits alongside 5-fold and 10-fold cross validation. Results show consistent improvements over single and dual stream baseline models, with mean accuracy reaching 99.54% in 10-fold cross validation. Our framework offers a robust and interpretable approach to MRI-based brain tumour classification by integrating complementary feature fusion with clinically responsible explainability.
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Manuscript Title:

A Trimodal Feature Fusion Framework for Explainable Brain Tumour Classification Using MRI Images

Journal: *Computer Methods and Programs in Biomedicine*

Dear Editor-in-Chief,

We are pleased to submit our manuscript entitled **"A Trimodal Feature Fusion Framework for Explainable Brain Tumour Classification Using MRI Images"** for consideration for publication in *Computer Methods and Programs in Biomedicine*.

This paper proposes a trimodal feature fusion framework that integrates convolutional neural networks and transformer-based representations for accurate and explainable brain tumour classification using MRI images. A multi-phase training strategy with exponential moving average refinement is employed to improve model generalisation, and a structured Grad-CAM based explainability framework is incorporated to support clinically meaningful interpretation.

The proposed work aligns well with the aims and scope of *Computer Methods and Programs in Biomedicine*, particularly in medical image analysis and decision support systems. The manuscript presents original work that has not been published previously and is not under consideration elsewhere, and all authors have approved its submission.

Thank you for your consideration.

Yours sincerely,
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Highlights

- A trimodal feature fusion model that brings together CNNs and transformer networks to classify brain tumours from MRI images.
- A multi-phase training approach that uses exponential moving average refinement to help the model generalise better.
- A structured explainability framework built on Grad-CAM that keeps visualisations focused and useful for clinical decision making.

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A Trimodal Feature Fusion Framework for Explainable Brain Tumour Classification Using MRI Images

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Abstract

Accurate classification of brain tumours from magnetic resonance images is critical for computer-aided diagnosis, enabling precise tumour typing that supports timely clinical assessment and treatment decisions. The challenge lies in the substantial heterogeneity of tumours and the scarcity of high quality, well annotated medical imaging datasets. Most state of the art approaches employ deep learning models, particularly convolutional neural networks and their variants, typically organised as single or dual stream systems optimised for classification accuracy. Many existing methods also rely on unconstrained post-hoc visualisations for explainability, which can yield ambiguous or clinically unreliable interpretations. In this work, we introduce a trimodal feature fusion framework for explainable brain tumour classification that combines a ResNet50-based CNN, a hierarchical transformer, and a lightweight vision transformer. A multi-phase training strategy with exponential moving average parameter aggregation is employed to enhance training stability and generalisation. Model interpretability is achieved through a structured Grad-CAM-based explanation protocol that quantifies activation strength while explicitly disclaiming any claim to tumour localisation or segmentation. We evaluate the framework using 70/30 and 80/20 train–test splits alongside 5-fold and 10-fold cross validation. Results show consistent improvements over single and dual stream baseline models, with mean accuracy reaching 99.54% in 10-fold cross validation. Our framework offers a robust and interpretable approach to MRI-based brain tumour classification by integrating complementary feature fusion with clinically responsible explainability.

Keywords: Brain tumour classification, Magnetic resonance imaging (MRI), Trimodal feature fusion, Deep learning, Explainable artificial intelligence, Grad-CAM

1. Introduction

Brain tumours are among the most dangerous neurological conditions, causing high rates of illness and death when not diagnosed and treated early. When abnormal cells grow uncontrollably in the brain, they can severely disrupt normal brain function, which is why getting an accurate diagnosis is so important for choosing the right treatment. MRI has become the go to imaging technique for diagnosing brain tumours because it provides excellent soft tissue contrast, can capture images from multiple angles, and doesn't expose patients to ionising radiation [1, 2]. Even with today's high-resolution MRI scans, analysing them manually is far from straightforward. Brain tumours vary widely in their shape, size, texture, and how they appear in terms of intensity, even when comparing tumours of the same type. A meningioma, for instance, often exhibits similar intensity profiles on MRI scans, complicating accurate classification [3, 4]. Manual interpretation is not only time intensive but also prone to variability between observers, which has motivated the push toward automated diagnostic systems [5, 1]. In recent years, deep learning has emerged as a promising tool for medical image analysis, with CNN proving particularly effective for brain tumour diagnosis [6, 4]. These networks can extract relevant spatial features directly from image data without extensive manual preprocessing [6, 2]. That said, standard CNN models are inherently limited by their local receptive fields, often failing to capture the broader contextual patterns that distinguish different tumour classes [7, 8]. To solve this, researchers have been building hybrid models that combine CNNs with transformers or ensemble techniques, allowing them to learn local and global features together [7, 8]. Transformers, which were first developed for understanding language, turn out to be good at capturing long range relationships in images through their self-attention mechanisms. This complements what CNN does well, which

is picking up on local features [9, 8]. Meanwhile, there's also been growing interest in making these models more interpretable using explainable AI techniques like Grad-CAM and attention maps [10, 7]. Still, there are gaps that need to be filled. A lot of the work so far has focused on simpler binary classification problems or has used fairly small datasets, which raises questions about whether the results will hold up in different clinical settings [6, 2]. Some models perform well but are essentially black boxes, which is a major problem when you're trying to use them in clinical practice [10, 7]. Others are so computationally demanding that they're not realistic for everyday use [11, 12]. On top of that, not many studies have really tested how robust these models are using different validation approaches like train-test splits and k-fold cross validation. We developed our trimodal brain tumour classification system to address these issues. The system takes advantage of complementary feature learning to improve both its ability to distinguish between tumour types and its generalisation to new data. We tested it extensively using multiple experimental setups and compared it against standard CNN models as well as CNN+Swin hybrid models. By combining local, global, and contextual features, our approach provides a practical way to classify brain tumours into multiple categories. Here's what we contribute:

- A trimodal feature fusion model that brings together CNNs and transformer networks to classify brain tumours from MRI images.
- A multi-phase training approach that uses exponential moving average refinement to help the model generalise better.
- A structured explainability framework built on Grad-CAM that keeps visualisations focused and useful for clinical decision making.

The rest of the paper is organised as follows. In section 2, a thorough overview of the literature is provided. Section 3 outlines the primary strategy for the proposed methodology. Section 4 describes the results obtained on the benchmark dataset. Section 5 covers the conclusion with future work.

2. Literature survey

Brain tumour classification using MRI has drawn significant research interest because of its importance for early diagnosis and treatment decisions. Early work in this area relied heavily on conventional machine learning paired with hand crafted features to identify abnormal tissue. While these methods produced reasonable results, they struggled with noise, required careful feature engineering, and didn't generalise well across different datasets [1, 13]. Traditional computer-aided diagnostic systems used texture-based features, statistical descriptors, and simple classifiers to detect tumours. The introduction of CNN marked a major shift, as these networks could learn features automatically from MRI scans without manual intervention. Early CNN models showed that deep architectures were effective at capturing tumour characteristics and consistently outperformed older methods [2, 14]. As deep learning advanced, researchers moved toward end to end CNN models for classifying multiple tumour types. These models could distinguish between glioma, meningioma, and pituitary tumours when trained on raw MRI data, though they often struggled with limited training samples and tended to overfit [6, 4]. Transfer learning became a popular solution to the data scarcity problem. Architectures like ResNet, VGG, and EfficientNet, which were originally trained on large scale natural image datasets, have been successfully adapted for medical imaging tasks. Fine-tuning these pretrained models improved both training speed and classification accuracy, demonstrating their value even with smaller MRI datasets [15, 16]. Several groups tried something different. They didn't use CNN for the whole classification pipeline; instead, they trained CNN just to pull out useful features from the images, then handed those features off to simpler algorithms like SVMs or KNN to do the classifying. The technique delivered solid results. Accuracy held up well, and the computational requirements were notably lower, both important factors when training datasets tend to be small [5, 17]. Transformers have more recently entered the picture as a way to overcome CNN's inherent focus on local regions. Vision Transformers and hybrid CNN + Transformer models have shown they can capture long range dependencies in MRI scans, which has improved classification for more complex tumour patterns [8, 18]. Attention mechanisms have also been integrated into deep learning pipelines to help models focus on diagnostically important regions. These attention-based models improved both accuracy and interpretability by highlighting relevant features, though their success depended heavily on the specific architecture and training approach used [7, 19]. Explainable AI has gained momentum in medical imaging, with Grad-CAM being one of the most widely used techniques. Grad-CAM allows researchers to see which regions of an image influence a CNN's

predictions, providing at least partial insight into the decision making process [10, 9]. Not all recent work has focused on deep learning. A few groups have applied mathematical methods like topological data analysis and tensor decomposition to find structural patterns in MRI data. These techniques provide a different way to approach classification, and they may offer clearer interpretability than standard deep learning methods [3, 20]. Interestingly, research focused on tumour segmentation has also yielded insights relevant to classification. Transformer-based segmentation models, for instance, have demonstrated how to effectively combine global contextual information with encoder-decoder frameworks, though they Ftend to be computationally expensive [21, 22]. Given the practical constraints of clinical deployment, several groups have focused on developing lightweight architectures. The goal is to simplify model design without losing classification accuracy. These approaches streamline transformer architectures and adjust normalisation techniques to work well with smaller MRI datasets, sidestepping the computational burden that larger models typically impose [23, 24]. CNN variants like DenseNet and EfficientNet have also been evaluated for brain tumour classification. Both architectures rely on dense connections and dropout mechanisms to encourage better feature reuse and limit overfitting, critical considerations given how limited medical imaging data tends to be [11, 25]. Lately, there's been growing interest in models that build explainability directly into their design. These hybrid architectures combine CNN's and transformers with visualisation tools like Grad-CAM, attempting to address the interpretability challenges that have long plagued deep learning system models [12, 26]. When tested on standard benchmarks, CNN-only models remain competitive, but hybrid and ensemble approaches tend to outperform them, especially where tumour types are visually similar [27, 2]. Multiple studies have emphasised the advantages of combining different learning strategies. Using multiple architectural approaches or integrating different imaging modalities can make models more robust, although ensuring that features from different sources work well together while maintaining interpretability has proven difficult [19, 28]. Great benchmarking efforts like BRATS have revealed just how variable and challenging brain tumour analysis can be. These benchmarks have shown substantial disagreement even among expert annotators, indicating the importance of systems that are both reliable and generalizable [20, 29]. Current trends point toward architectures that blend diversity with explainability to improve clinical usefulness. Hybrid CNN+Transformer models with built-in interpretability mechanisms have shown promise, reflecting a broader recognition that explainability isn't optional in clinical settings [8, 18]. Despite all this progress, most existing methods still depend on one or two architectural approaches, and many provide visualisations that lack proper constraints or training stability. These limitations point to the need for frameworks capable of integrating diverse architectures, robust optimisation, and responsible explainability for MRI-based brain tumour classification [7, 12].

3. Proposed methodology

The proposed approach is designed to carry out robust and interpretable classification of brain tumours from MRI images by effectively combining the most complementary feature representations, stabilised training, and constrained visual interpretability. It has three major components: a trimodal feature extraction structure, a multi-phase training strategy with EMA parameter aggregation, and a structured explainability protocol based on Grad-CAM. A conceptual overview of the proposed pipeline is shown in Figure 1.

3.1. Experimental setup

For all experiments, the deep learning platform used is PyTorch. The proposed trimodal model consists of a combination of a convolutional neural network and two transformer networks, all of which are trained using a multi-phase optimisation technique. Training, testing, and all computations are carried out using GPUs whenever they are available. Robust estimation is facilitated by using different evaluation protocols, such as various train–test splits and cross validation methods. The EMA of the model parameters is also maintained during the final phase of the training process. The single stream and dual stream architecture based baseline experiments are conducted under the same experimental setup to ensure a fair comparison with the proposed framework. The remaining two baseline models consist of a single stream CNN model and a two stream CNN + Swin Transformer model. These models are also trained and tested under the same experimental settings as the proposed trimodal framework. The preprocessing steps are identical, including the training and testing split ratios (70/30 and 80/20), as well as the cross validation settings (5-fold and 10-fold).

3.2. Dataset

These experiments have been conducted on publicly available data known as the Brain Tumour MRI

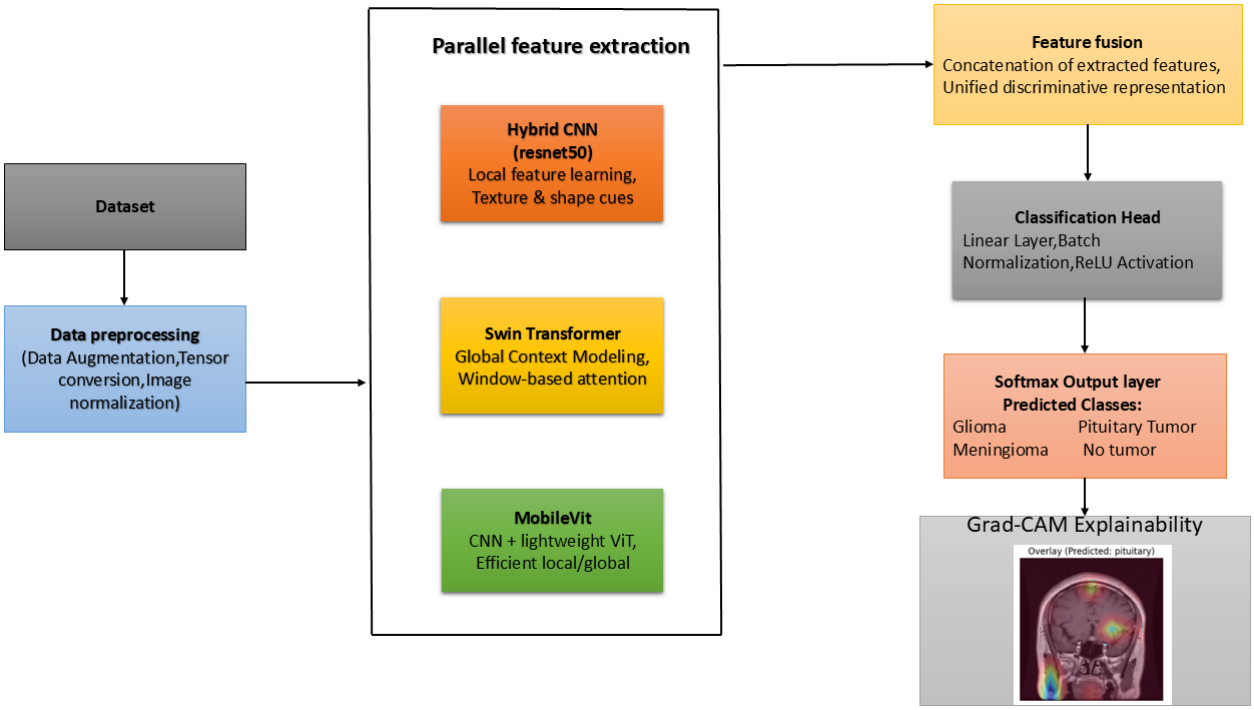


Figure 1: Architecture of the proposed methodology.

Dataset[30]. The dataset consists of a total of 7,023 MRI images belonging to four classes of brain tumours. These classes include glioma, meningioma, pituitary tumour, and no tumour (normal). These types of tumours are typical examples of clinical cases that are commonly encountered. They are highly variable in terms of their shapes, textures, intensities, and sizes. These factors contribute to the complexity of classification. Therefore, various feature representation schemes are necessary. The conditions for forming MRI images are not standardised. Variability is therefore introduced both within classes and between classes. It is therefore a very diverse dataset for testing the robustness of the proposed trimodal classification approach. In Figure 2, the class-wise representation of the images within the dataset is shown, and the images are evenly distributed among the four classes.

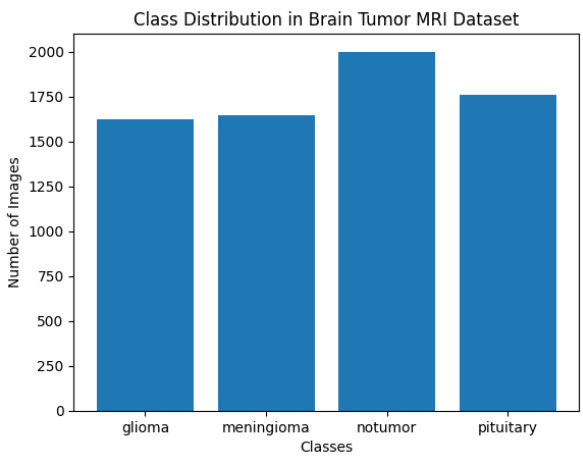


Figure 2: Class-wise distribution of MRI images across four tumour categories.

3.3. Preprocessing

All the MRI images go through a preprocessing pipeline before the models are trained, ensuring consistency in the learning process across different experimental configurations. The preprocessing techniques help make the input images suitable for the network backbones while eliminating variability caused by acquisition conditions. First, every MRI image is resized to a fixed resolution of 224×224 pixels to ensure that all images are aligned with the required inputs for all the pre-trained models used in the proposed framework. Resizing ensures that all images have uniform dimensions for effective transfer learning. Then, the pixel intensity values are normalised based on the mean and standard deviation values of each channel, which are derived from

the ImageNet dataset. In this way, the intensity values of MRI images follow a distribution on which models are already trained, hence ensuring stable gradient updates during training, leading to improved convergence. Data augmentation is used as a means of improving generalisation and fighting overfitting during training. The techniques include random horizontal flipping, small random rotations, and slight colour variation. These processes introduce variability with regard to position and appearance. However, none of these are applied during the validation and testing phases to ensure there is no bias. The same set of preprocessing techniques is applied to the proposed trimodal approach and the baseline single-stream and dual-stream models. In this way, any disparities in model performance will solely depend on the differences among the model's architectures and not on the preprocessing techniques applied to the data.

3.3.1. Splitting dataset

There are several dataset splitting methods that are adopted to comprehensively assess the robustness and generalisation ability of the proposed framework. They are designed to investigate the performance of models with varying training settings and to provide a fair comparison between the proposed and existing models. First, fixed train-test splits are used, with 70/30 and 80/20 splits of the dataset into training and testing data, respectively. This approach evaluates classification directly on unseen data and takes into consideration common practices in brain tumour classification. Besides the fixed splits, 5-fold and 10-fold cross validation techniques are also considered to further the stability and consistency of the model. In this case, the data are split equally, and each subset is used as a validation set for one iteration while the remaining subsets are used for training. All performance metrics are then averaged to counteract data partitioning biases. Stratified sampling is applied in all the splitting approaches to ensure that there is equality in class distribution in both training and testing data. This is done in both the proposed model and the baseline models.

3.3.2. Multi-Phase Training Strategy

A multi-phase optimisation strategy is utilised to enhance training stability and generalisation. Only the classification head is trained, while all the feature extraction backbones are frozen in the first stage of training. This warm up can allow the classifier to adapt to the fused representation space without introducing unstable gradient updates to the pre trained backbones. In the second stage, all network parameters will be unfrozen and jointly fine tuned at a reduced learning rate, which is set to harmonise the three branch representations. Training continues at an even lower learning rate in the third phase while maintaining an exponential moving average of the model parameters. This EMA model is used during evaluation to reduce variance and further improve robustness across different validation settings.

3.4. Trimodal Feature Extraction

The proposed framework features three parallel pathways of feature extraction that capture diverse and complementary representations from MRI data. Each pathway operates on the same input image but with different emphases on its representation. The first pathway uses a ResNet50 convolutional neural network to extract fine-grained local spatial features such as textures, edges, and intensity variations, which are critical for tumour characterisation. In the second pathway, a hierarchical transformer, Swin Transformer, is used to model regional and hierarchical contextual relationships using window-based self-attention mechanisms. The third pathway uses a lightweight vision transformer, MobileViT, which integrates convolution operations with transformer blocks to capture global contextual information in an effective manner. The feature vectors from the three pathways are combined at the feature level to form an integrated representation. This fusion strategy maintains the discriminative information of each backbone without imposing decision-level constraints and thus allows the classifier to exploit complementary spatial and contextual cues.

3.5. Feature selection approaches

In this proposed framework, there are no explicit manual procedures for feature selection. This is because architectural diversity enables implicit feature learning from MRI images. There are three backbone architectures for feature extraction to better capture tumour characteristics. The ResNet50-based CNN is designed to learn local fine grained features concerning edges, textures, and intensity information. The Swin Transformer is equipped with a window-based self-attention mechanism to capture regional and hierarchical contextual relationships, thus effectively capturing mid level context information. The MobileViT is a hybrid convolutional-transformer model that combines lightweight transformers to capture global contextual information. Feature concatenation is carried out to create a unified representation based on the feature vectors obtained from each of the branches. In feature-level fusion, no prior restriction is imposed on each branch to

make an independent classification decision. The classifier is provided with an enhanced feature representation by combining local, regional, and global information into a unified representation. In this representation-based design, there is no manual selection of features, and the feature usage remains constant for all experimental configurations. The same technique of feature fusion is also used for the proposed trimodal model and the baseline single stream and dual stream models, thus ensuring a proper comparison of performance.

3.6. Classification Head

This merged feature representation is fed into a fully connected classification head, which consists of a dimensionality reduction step, batch normalisation, non-linear activation, and dropout regularisation in order to avoid overfitting and allow for stable learning when training on limited medical imaging data. The last layer outputs class logits for the tumour classes.

3.7. Explainability and Structured Explanation Card

Visual explanations are obtained using Grad-CAM from the convolutional branch of the proposed network framework. The final convolutional layer explanation is generated from Grad-CAM to reflect the regions that most strongly contribute to the predicted class. Instead of presenting such visual explanations in isolation, an SEC is introduced to constrain interpretation and provide additional context. Every SEC aggregates the predicted class, confidence score, quantitative confidence gap between top predictions, and Grad-CAM activation strength into an explanatory representation. This card explicitly indicates that highlighted regions correspond to discriminative features used for classification and do not correspond to anatomical tumour localisation or segmentation.

3.8. Methodological Scope

The proposed approach focuses on image level classification and explainability only, no tumour segmentation or pixel level diagnostic conclusion is made. This clear scope thus defines that the model outputs are within clinically appropriate use and discourages over-interpretation of the visual explanations.

4. Results and Discussion

The framework for MRI brain tumour classification has been built upon the concept of trimodal feature fusion as described within this section, with comprehensive testing of all aspects of model classification performance (including accuracy, precision, recall, and F1-score), along with an assessment of the stability of the proposed model across different evaluation configurations. The experimental design tests and validates the proposed trimodal model by creating multiple train/test partitions (70/30, 80/20), as well as performing five-fold and ten-fold cross validation on these partitions. Each model has been evaluated under identical conditions for preprocessing, dataset partitioning, and evaluation metrics to eliminate bias introduced by experimental design choices. The testing and validation were focused specifically on classification performance under different architectural configurations. The results indicate that architectural diversity improved classification performance and that feature level fusion further contributed to this improvement. Along with the quantitative metrics, we also used confusion matrices and classification reports for each of the four classes to determine the performance of each class, as well as to analyse error distributions for each of the tumour classes. In particular, scores for recall and F1-score are highlighted because they have the highest clinical relevance for medical diagnosis of brain tumours. In addition, the effectiveness of the multi-phase training method implemented using EMA aggregation has been evaluated with respect to robustness and consistency across different validation settings. Lastly, this section continues to offer structured instances of explainability using Grad-CAM-based visual analysis, with a reminder that explanatory outputs must only demonstrate interpretability and, therefore, must not imply tumour localisation or segmentation. As such, the results demonstrate that the trimodal framework provides a coherent basis for utilising each of the evaluation methods and confirms the advantages of complementary features through using a stabilised training environment, enhancing interpretability for classifying MRI brain tumours. The Figure 3 represents the comparison of the baseline and proposed model.

4.1. Performance evaluation metrics

The trimodal framework for classifying brain tumour images has been assessed using a variety of performance metrics to measure effectiveness. In particular, these metrics include accuracy, precision, recall, and F1-score. The reason for including all four metrics was to provide insight into both the overall performance of the classification model and its reliability within each individual class of tumour, which is critical when classifying multi-class medical imaging, where class type may have different clinical implications. Accuracy gives a general indication of performance across all tumour classes by measuring the ratio of correctly classified samples to total samples. When evaluating

Table 1: Overall accuracy comparison of different models under various training and validation strategies

Model	**70/30**	**80/20**	**5-Fold (± std)**	**10-Fold (± std)**
CNN	94.26	92.95	93.83 ± 1.59	94.74 ± 0.52
CNN+Swin Hybrid model	98.81	98.36	98.99 ± 0.72	99.18 ± 0.59
Proposed Trimodal	98.77	99.36	99.10 ± 0.30	99.54 ± 0.30

Table 2: Class-wise precision, recall, and F1-score of the proposed trimodal model under different evaluation settings

Proposed 70/30 Split				**Proposed 80/20 Split**		
Class	**Precision**	**Recall**	**F1-score**	**Precision**	**Recall**	**F1-score**
Glioma	99.59	98.97	99.28	100.00	99.07	99.53
Meningioma	98.75	96.15	97.44	99.39	99.09	99.24
No Tumour	97.56	99.83	98.68	99.50	99.75	99.63
Pituitary	99.43	99.81	99.62	99.15	100.00	99.58
Proposed 5-Fold				**Proposed 10-Fold**		
Class	**Precision**	**Recall**	**F1-score**	**Precision**	**Recall**	**F1-score**
Glioma	99.69	100.00	99.85	100.00	100.00	100.00
Meningioma	99.39	98.48	98.93	99.94	99.21	99.57
No Tumour	99.50	100.00	99.75	99.75	99.60	99.57
Pituitary	99.43	99.43	99.43	99.10	99.94	99.57

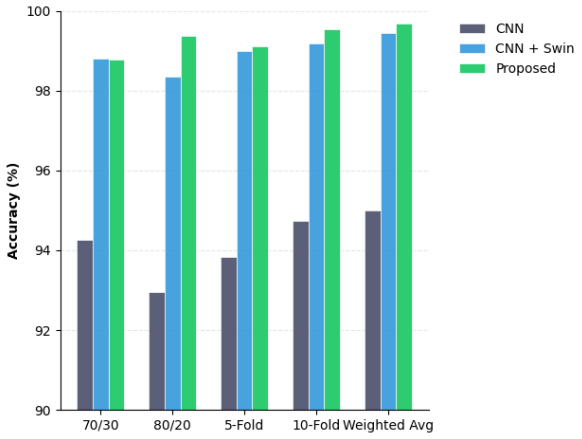


Figure 3: Accuracy comparison of CNN, CNN+Swin, and the proposed model under different train–test splits and cross validation strategies.

the accuracy against two related systems (the baseline CNN and the baseline CNN + Swin), the relative accuracy can be assessed directly, particularly because the sample sets were relatively class-balanced. In addition to accuracy, several other measures can be used to evaluate the ability of a model to accurately predict clinical outcomes. One such measure is the precision and recall scores for each tumour type. The precision of a model measures how well a model classifies a tumour type correctly by indicating the percentage of predicted cases that were correctly classified versus the total number of predictions made for that tumour type. A high precision score means that the model is producing very few false positives, indicating that the model does a good job of not misclassifying tumours. It is important to avoid false positives in the classification of brain tumours because misclassifying a brain tumour will result in incorrect treatment options. Recall is also referred to as sensitivity, and it is a measurement of a model's effectiveness at correctly identifying all actual tumours of a particular type in the dataset. This measurement becomes even more important in the medical field because if a tumour is missed, the results of treatment may be dire for the patient. The proposed trimodal classification model increases recall performance by using feature representations that incorporate complementary local and global information into the overall MRI image. The F1-score, the harmonic mean of precision and recall, can serve as a fair measure of the amount of false negatives and false positives present in an evaluation. It is important for the evaluation of the proposed method because the class labels can vary greatly in terms of the nature, size, and intensity distribution of the classes. Obtaining an

F1-score prevents methods from being biased toward obvious classes. To evaluate the robustness and generalizability of a method, the evaluation metrics are calculated for each class and evaluated through multiple experimental configurations. The evaluation of methods is based on the hold out validation technique (70/30 and 80/20 splits) and k-fold cross validation (5-fold and 10-fold). This allows researchers to ensure that the reproducibility of the trimodal model is not dependent on a specific data split.

4.2. Experimental Results and Comparative Analysis

The experimental results derived from the proposed trimodal brain tumour classification framework have been analysed against a baseline CNN model and a CNN+Swin hybrid model. Performance has been assessed using multiple configurations, including k-fold (5-fold and 10-fold) cross validation and hold-out (70/30 and 80/20) train-test splits. A comprehensive view is taken of the performance of the models through an assessment of general accuracy as well as detailed analysis at the per-class level. The overall accuracy comparison of the proposed and baseline models under different evaluation strategies is summarised in Table 1.

4.3. Overall Performance Comparison

When analysed across multiple configurations, the overall classification accuracy of the various models presents considerable differences in performance levels. The baseline CNN model was shown to achieve moderate general accuracy at all configurations, indicating that it could effectively learn discernible features from an MRI image and subsequently improve its overall performance; however, it still had significant limitations in its ability to effectively learn how to differentiate between the many complex variations of brain tumours. As shown in Table 1, the proposed trimodal model consistently outperforms the baseline CNN and CNN+Swin hybrid models across all evaluation strategies. By combining convolutional and transformer-based feature learning, the CNN+Swin hybrid model consistently outperformed the baseline CNN. The hybridisation of these two unique styles of feature learning takes both the image's local texture features and its global context and association, and subsequently processes the information gathered to develop a clearly defined improvement over all of the previously validated models. The proposed model achieves the highest accuracy across all of the experimental configurations tested. The trimodal model exhibits higher performance than the baseline models based on accuracy when using either of the two splits, 70/30 or 80/20; however, it is also evident from the 5-fold and 10-fold cross validation experiments conducted that this particular model demonstrates consistently high accuracy with a low degree of variation across the folds. This indicates that the accuracy gain of the trimodal model does not rely on a specific split of the data.

4.4. Class-wise Performance Analysis

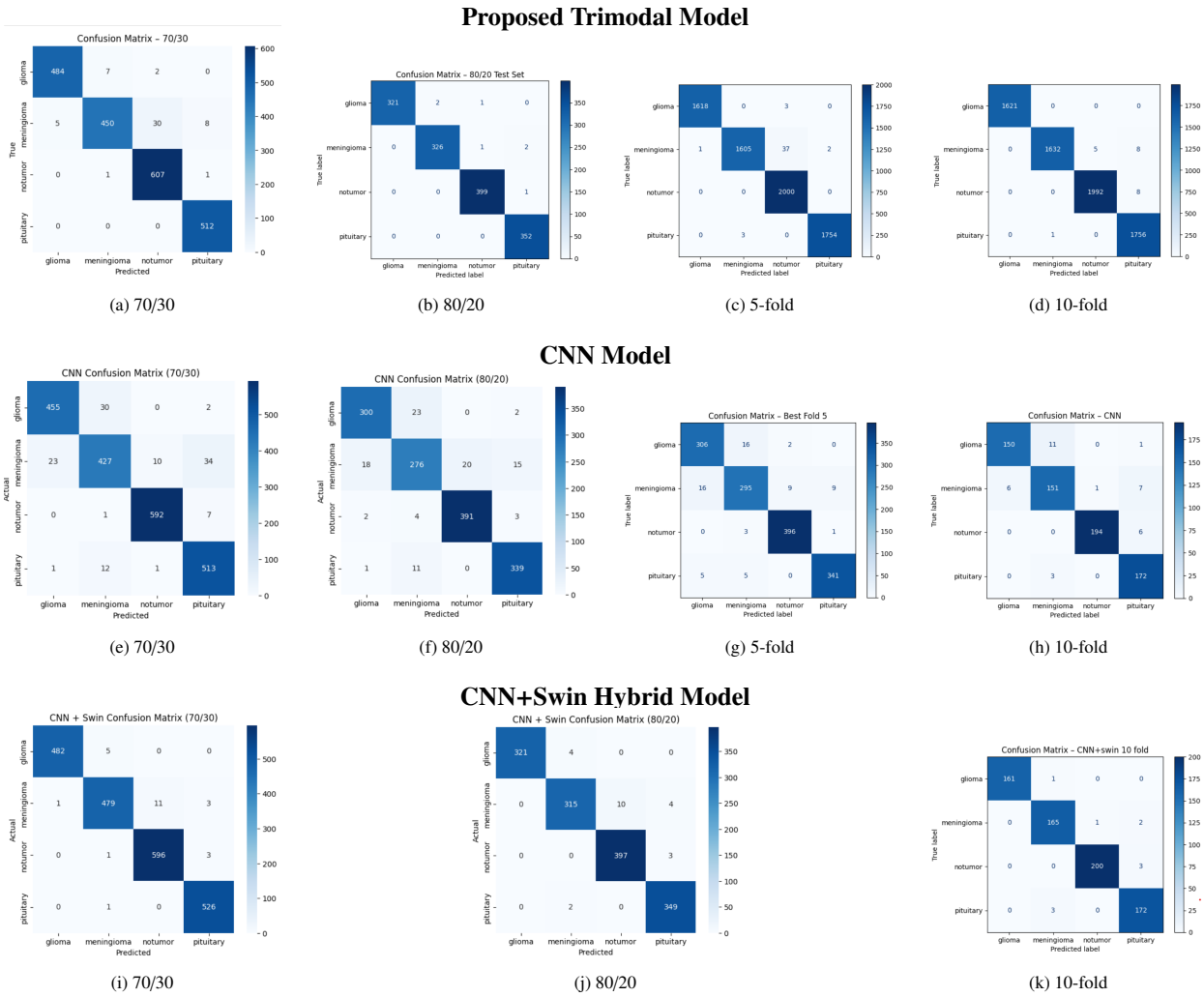
The performance of the proposed trimodal model is clearly supported by a detailed analysis of model performance at the class level. The model has demonstrated a consistently strong performance across all experimental configurations, achieving high amounts of precision, recall, and F1-scores in every case for all four types of tumours: glioma, meningioma, pituitary tumour, and no tumour. The proposed model is particularly successful for the tumour types that are often difficult to separate from each other because of their shared visual characteristics (glioma and meningioma are representative examples). High recall values indicate that this model has very few false negatives. Similarly, high precision values indicate that this model has very few false positives. The ability to minimise false negatives and false positives is of utmost importance in this type of application (medical imaging) because either of these two possibilities can lead to significant clinical outcomes. The trimodal architecture has significantly reduced misclassification in all categories compared with CNN (the baseline model). While the misclassification gap has been closed to some extent by the CNN+Swin hybrid model, the proposed architecture should still provide superior misclassification performance, especially for cross validation experiments. The findings indicate the advantage of multiple feature extraction methods being merged into a single model. The class-wise precision, recall, and F1-score values of the proposed trimodal model under different evaluation settings are summarised in Table 2.

4.5. cross validation and robustness

From the k-fold cross validation results, the strength of this model's generalisation ability is demonstrated. The trimodal model was stable over different cross validation folds in terms of model performance based on 5-fold and 10-fold cross validation and had little change between each fold with regard to accuracy and performance in each class.Based on this evidence, the proposed trimodal architecture will generalise well to other data that were not included in the training/testing sets and is expected to have little to no sensitivity to how training/testing partitions were divided. Therefore, the

Table 3: Class-wise precision, recall, and F1-score of CNN and CNN+Swin hybrid models under different evaluation strategies

CNN Model												
Class	**70/30 Split**			**80/20 Split**			**5-Fold**			**10-Fold**		
	Precision	**Recall**	**F1-score**	**Precision**	**Recall**	**F1-score**	**Precision**	**Recall**	**F1-score**	**Precision**	**Recall**	**F1-score**
Glioma	95.00	93.00	94.00	93.00	92.00	93.00	93.58	94.44	94.01	96.15	92.59	94.33
Meningioma	91.00	86.00	89.00	88.00	84.00	86.00	92.48	89.67	91.05	91.51	91.51	91.51
No Tumour	98.00	99.00	98.00	95.00	98.00	96.00	97.30	99.00	98.14	99.48	97.00	98.22
Pituitary	92.00	97.00	95.00	94.00	97.00	95.00	97.15	97.15	97.15	92.47	98.28	95.29
CNN+Swin Hybrid Model												
Class	**70/30 Split**			**80/20 Split**			**5-Fold**			**10-Fold**		
	Precision	**Recall**	**F1-score**	**Precision**	**Recall**	**F1-score**	**Precision**	**Recall**	**F1-score**	**Precision**	**Recall**	**F1-score**
Glioma	100.00	99.00	99.00	100.00	99.00	99.00	98.62	97.62	97.65	100.00	99.38	99.69
Meningioma	99.00	97.00	98.00	98.00	96.00	97.00	97.57	98.54	98.77	99.40	99.21	99.20
No Tumour	98.00	99.00	99.00	98.00	99.00	98.00	99.00	99.31	99.12	99.43	99.47	99.54
Pituitary	99.00	99.21	99.00	98.00	99.00	99.00	98.58	98.97	99.01	99.00	99.21	99.32

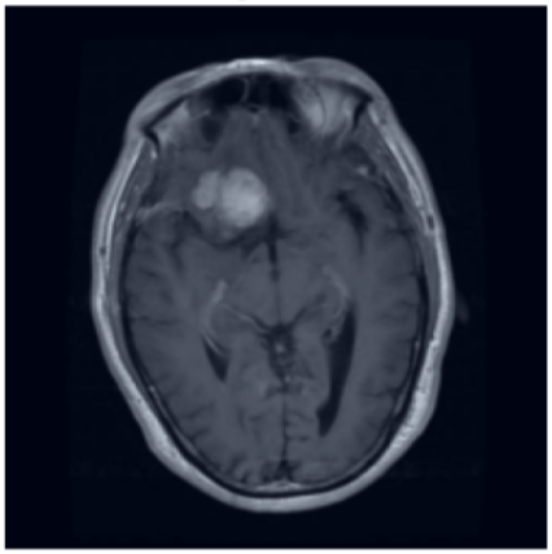


(a) 70/30 (b) 80/20 (c) 5-fold (d) 10-fold

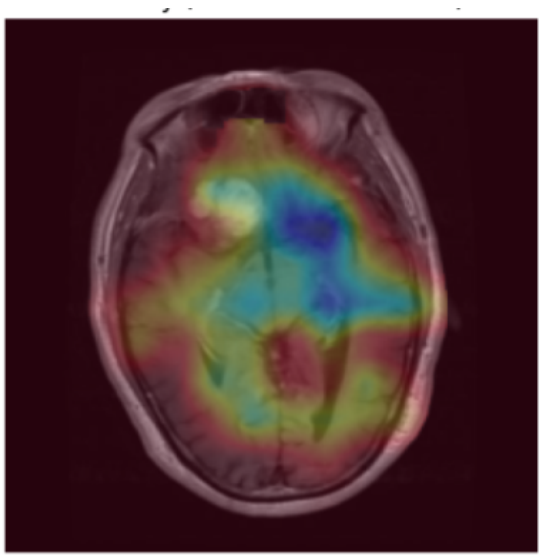
(e) 70/30 (f) 80/20 (g) 5-fold (h) 10-fold

(i) 70/30 (j) 80/20 (k) 10-fold

Figure 4: Confusion matrices of the proposed trimodal, CNN baseline, and CNN–Swin hybrid models under different evaluation strategies



(a) Input MRI image



(b) Grad-CAM overlay

	SEC Section	Attribute	Value
0	Prediction Summary	Predicted Class	Meningioma
1	Prediction Summary	Predicted Index	1
2	Confidence Analysis	Prediction Confidence	0.999863
3	Confidence Analysis	Top-1 vs Top-2 Margin	0.999791
4	Attention Evidence (Grad-CAM)	Mean Attention Strength	0.099084
5	Attention Evidence (Grad-CAM)	Max Attention Strength	1.0
6	Attention Evidence (Grad-CAM)	Active Attention Area Ratio	0.059451
7	Decision Reliability	High Confidence Decision	True
8	Decision Reliability	Clear Class Separation	True

(c) Structured Explanation Card (SEC)

Figure 5: Explainability and decision reliability illustration of the proposed trimodal brain tumour classification framework, showing the input MRI image, Grad-CAM visualisation, and the corresponding Structured Explanation Card (SEC).

model can be considered suitable for real-world applications, where data distributions can vary depending on the clinical environment.Table 3 reports the class-wise precision, recall, and F1-score of CNN and CNN+Swin hybrid models across different evaluation strategies. All confusion matrices associated with the various experimental configurations indicate a consistent classification behaviour of the trimodal model. Across all validation methods, the classification patterns observed in the confusion matrices exhibit a high diagonal concentration, indicating a substantial number of correctly predicted samples for each tumour class. The discriminatory ability of the trimodal model is further supported by the low level of misclassification among visually similar tumour subclasses. Taken together, these observations are consistent with the high precision and recall values reported in the quantitative evaluation. Figure 4 presents the confusion matrices of the proposed trimodal model, the CNN baseline, and the CNN + Swin hybrid model under different evaluation strategies.

Table 4: Comparison of Brain Tumour Classification Methods with Reported Performance Metrics

S.No	Authors	Year	Methodology	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
1	Seetha and Raja [2]	2018	CNN with small kernels	97.50	–	–	–
2	Sultan et al. [4]	2019	Deep CNN for multi-class classification	96.13	96.10	96.10	96.10
3	Jia and Chen [21]	2020	FAHS–SVM framework	98.51	–	–	–
4	Divya et al. [26]	2020	ResNet50 transfer learning	98.67	–	–	–
5	Saleh et al. [15]	2020	CNN-based transfer learning	98.75	–	–	–
6	Biswas and Islam [17]	2021	Hybrid CNN–SVM model	98.55	–	–	–
7	Majib et al. [31]	2021	VGG-SCNet stacked classifier	99.20	99.20	99.10	99.20
8	Nallamolu et al. [27]	2022	CNN-based multi-class classification	94.72	–	–	–
9	Vankdothu and Hameed [14]	2022	RCNN with GLCM features	95.17	–	–	–
10	Nayak et al. [16]	2022	Dense EfficientNet CNN	98.78	–	–	–
11	Sujatha and Rao [25]	2023	DenseNet201-based CNN	88.00	–	–	–
12	Hossain et al. [8]	2023	Vision Transformer ensemble	96.94	–	–	–
13	Ferdous et al. [23]	2023	LCDEiT vision transformer	98.11	–	–	–
14	De Benedictis et al. [3]	2024	TDA with Tucker decomposition	97.28	–	–	–
15	Guluwadi et al. [10]	2024	ResNet50 with Grad-CAM	98.52	98.50	98.50	98.50
16	Abdelhaliem et al. [19]	2025	Multimodal attention network	92.86	92.86	80.00	85.90
17	Dintakurthi et al. [18]	2025	CNN + Transformer hybrid	97.80	–	–	–
18	Hasan et al. [7]	2026	Attention-based explainable DL	97.00	97.00	97.00	97.00
19	**Proposed HybridTriModel**	**2026**	**ResNet50 + Swin Transformer + MobileViT**	**99.54**	**99.69**	**99.69**	**99.69**

4.6. Explainability Results

To ensure the Trimodal Brain Tumour Classification Framework is transparent and clinically interpretable, we provide an explanation and analysis of the reliability of this framework for clinicians and radiologists in making clinical decisions. This includes integrating a Grad-CAM assessment of the classification and a Structured Explanation Card (SEC). Figure 5 shows an example of the data that goes into this analysis, including the baseline MRI scan, the Grad-CAM overlay, and the associated SEC. The Grad-CAM overlay shows which region of the MRI scan had the strongest influence on the classification prediction.

From the overlay, the highlighted regions indicate areas that contribute most to the classification decision. This suggests that the model focuses on clinically relevant regions rather than background artifacts that are irrelevant to the classification process. The regionally limited and high concentration of activation sites indicates that the trained model has learned to discriminate based on features of the tumour, and therefore, it provides supporting evidence that the model's decisions are reliable. These activation maps represent model attention and should not be interpreted as explicit tumour localisation or segmentation. The Structured Explanation Card extends the qualitative insights provided by Grad-CAM. It also provides information on the measures related to the confidence of prediction and attention behaviour. For example, the top 1 vs top 2 prediction probability margin and the prediction confidence report show the level of distinction between the two classes that are competing. This means that the decision boundary of the model is stable and well defined. Additionally included are attention-related metrics, such as the average attention strength, the maximum attention strength assigned to a particular area, and the ratio of the image area receiving active attention relative to the entire image. Together, these metrics give an understanding of both how intensely and where the model is focusing. Based on lower active attention area ratios compared to the much larger values for peak attention, it appears that most of the model's attention is focused on more compact and salient areas, as opposed to being spread across a large portion of an image. The decision reliability measures derived from the metrics above validate the degree of confidence displayed in the prediction. Strong confidence and well defined separation of classes show that the model generates both confident and unambiguous results for this sample. Decision reliability indicators are important when applied

to medical imaging, as the clinical interpretation resulting from a lack of reliability or an unreliable prediction may lead to an incorrect clinical decision. Together, the use of Grad-CAM visualisation and structured explanation cards supports the assertion that the developed trimodal framework can create interpretable and trustworthy predictions in addition to achieving high classification accuracy. Performance, explainability, and confidence consistency are equally critical for clinical decision support systems; thus, by combining these characteristics into one framework, it is possible to increase the potential of this methodology for use in clinical decision making. The experimental results show that the proposed trimodal framework is more accurate, robust, and reliable than the baseline CNN and CNN+Swin hybrid models. The complementary nature of the features, together with the thorough evaluation and analysis of the framework, makes it possible to achieve high performance and clinical interpretability. These qualities make the proposed framework a strong candidate for a real world multi class brain tumour classification task using MRI images. Consistent performance across several splits and folds indicates that complementary feature learning, rather than memorisation, is responsible for the gains, despite the high accuracy achieved. Table 4 compares brain tumour classification methods from the literature, including studies that report only accuracy as well as those providing multiple performance metrics. The proposed HybridTriModel achieves the highest accuracy of 99.54% under 10-fold cross validation while maintaining strong and balanced performance across all evaluation metrics, demonstrating its effectiveness for multi-class brain tumour classification.

5. Conclusion

A trimodal feature fusion strategy for the classification of brain tumours from magnetic resonance imaging is proposed. The proposed framework combines a convolutional neural network with two transformer models. The strategy is capable of capturing local, regional, and global features from the MRI images. The feature fusion mechanism combines these features into a single embedding with improved discriminability, without being limited by decision level boundaries. Extensive experiments were performed using various evaluation methods, such as 70/30 and 80/20 train–test splits, as well as 5-fold and 10-fold cross validation. In all cases, the new model performed better than the single-stream and dual-stream baseline models, especially in terms of recall and the F1-score, which are important in medical image classification. The training process, involving multiple phases, together with the use of exponential moving average parameter aggregation, helped ensure robust optimisation. In addition to the quantitative performance, the framework also includes a structured explanation mechanism based on Grad-CAM. By integrating visual explanations into a Structured Explanation Card, the model enables transparent and interpretable results while deliberately refraining from making statements about tumour localisation or segmentation. This approach represents a form of constrained explainability.

In conclusion, the proposed trimodal approach shows that architectural diversity, stabilised training, and explainability control constitute an effective combination for brain tumour classification using MRI images. The findings suggest that the approach is suitable for image-level diagnostic decision support. Future work will aim to extend the proposed framework towards controlled tumour localisation and size estimation through the incorporation of segmentation aware learning modules, enabling more comprehensive spatial and volumetric analysis while maintaining interpretability and clinical reliability.

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Manuscript Title:

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This study does not involve any new experiments on human participants or animals. The MRI data used in this research were obtained from publicly available datasets, and all data were fully anonymised. Therefore, ethical approval and informed consent were not required for this study.